

 **DERCO CHILE**

BUSINESS INTELLIGENCE · FINAL INTEGRATIVE PROJECT · UDD 2026

From 550,033 car sales to one business decision.

Retail vehicle sales · May 2009 → March 2022

• Vicente Rodríguez · Agustín Reyes · Luis-Felipe Cáceres · Baptiste Vial

01 • THE CLIENT & THE QUESTION

DERCO hired us to find where the retail model leaks value.

“After 14 years of sales, where is DERCO leaking value — and what should change to grow margin?”

Decision-maker

DERCO's **Commercial Director** — owns brand mix, channel strategy and the marketing budget.

The decision

Where to move **commercial effort and budget** for FY2026.

Success

A ranked, **CLP-valued** list of moves the client can fund — not an accuracy score.

02 · THE DATASET & HOW WE WORKED

14 years of real transactions, through the full BI pipeline.

550K

transactions

464K

unique customers

9

brands · 1,543 models

511

comunas of Chile

1 · Frame

Question, decision-maker, 6 KPIs

2 · Prepare

Audit, clean, engineer. No leakage

3 · Model

Segment customers + predict losses

4 · Communicate

Insights, recommendation, dashboard

5 · Ethics

Bias, privacy, honest limits

Every row is one car sale — **date, customer hash, comuna, brand, model, channel, list price, margin**; two channels, **ces** (dealer, 65%) and **propio** (owned, 35%). One honesty note: the money fields are **synthetic / obfuscated**, so every CLP figure is a **relative signal** — shares, ranks and trends are valid, absolute pesos are not audited financials.

Python · pandas · scikit-learn · matplotlib · seed = 42 · one reproducible notebook — every number on every slide is exported by it into `metrics.json`.

03 · PREPARE & AUDIT — “CAN WE TRUST IT?”

Clean data first. Then we found a leak.

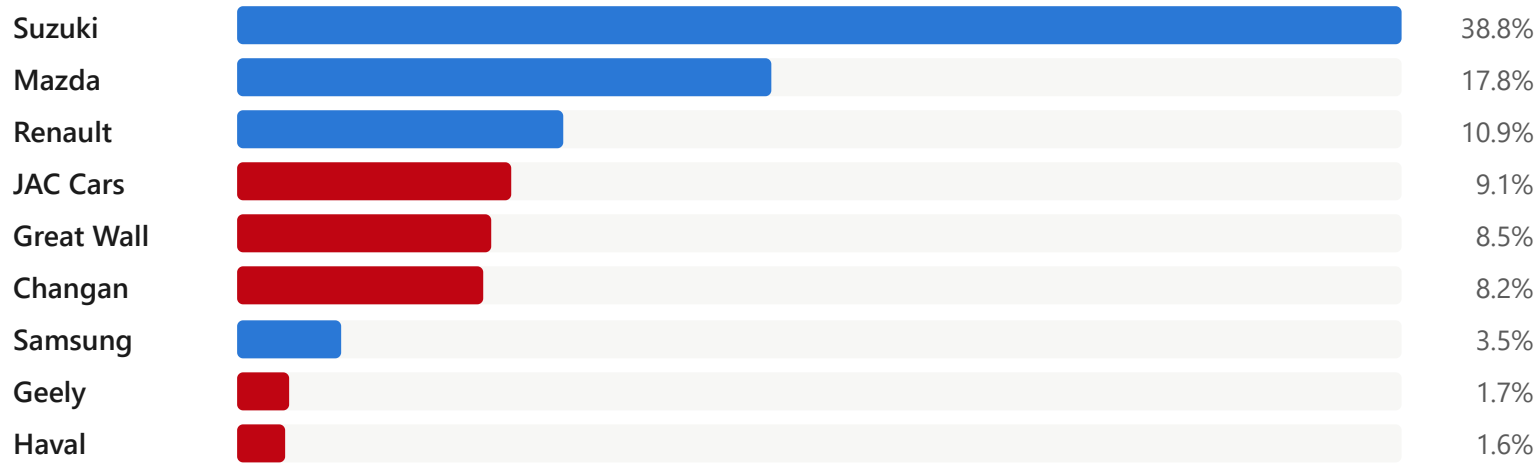
- <1% **missing**, zero duplicates, dates 100% consistent → safe to model.
- Missing comuna → flagged **UNKNOWN**, never dropped (dropping quietly shrinks revenue).
- Dropped the redundant **year** column; log-scaled skewed spend before clustering.
- **11,966 sales (2.18%) lose money** — sold below cost. A real finding, not a parsing error.

6.8bn

CLP of margin leaked through loss-making deals
≈ **2.7% of all retail margin** (synthetic money)

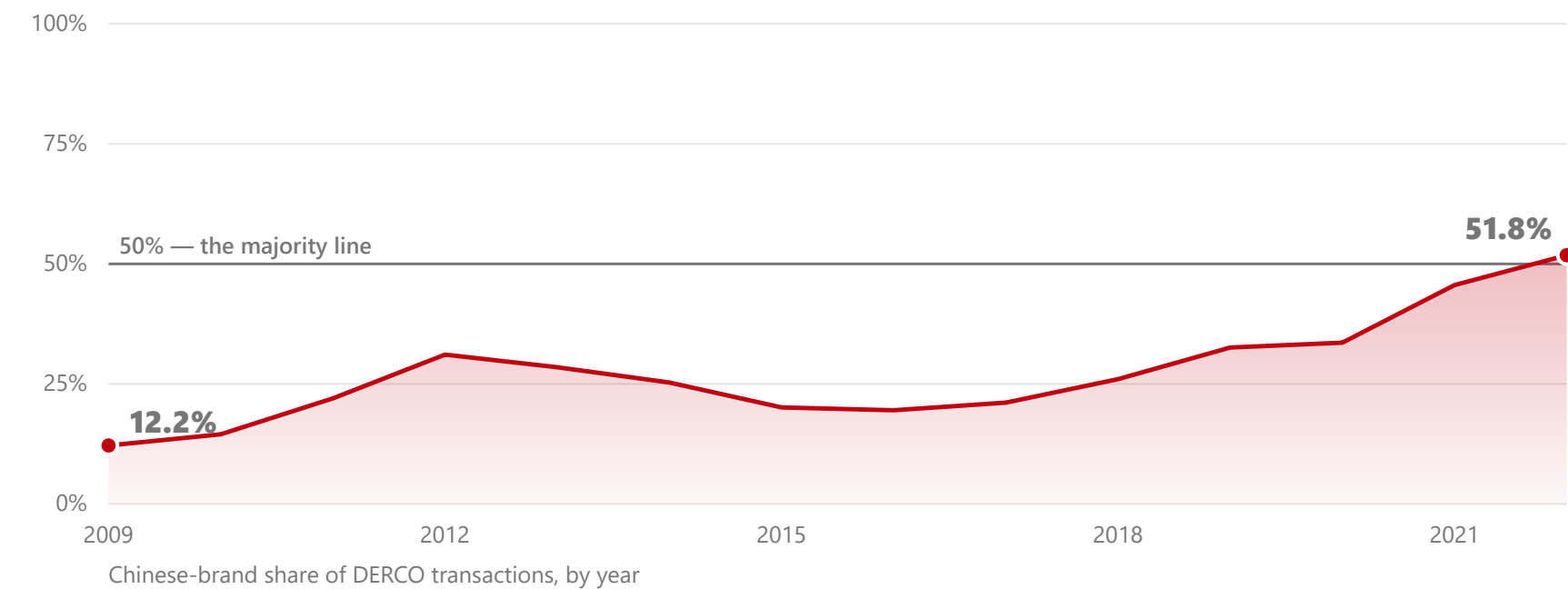
One brand carries the business.

■ Chinese brand ■ Incumbent brand



Share of all 550,032 transactions, 2009–2022. Bar length is volume; colour is brand origin — the split that matters strategically. **Suzuki alone is ~39%** → single-brand dependence risk.

China won the shelf.



Not a straight line — the share dipped to ~20% in 2015–16 before the 2019→2022 surge. But the direction is unambiguous: **Chinese brands now sell the majority of every car DERCOT moves.** 2009 (from May) and 2022 (to March) are partial years — shares are valid, totals are not.

Where, how, and when DERCO sells.

65 / 35

dealer **ces** vs owned **propio** — and dealers also price better: **6.2%** margin vs **5.2%**

33%

of all sales sit in the **top-10 comunas** — dense, but over-exposed to Santiago metro

Aug ▲

seasonal peak in **August** (December a close second), trough in **April** — plan stock & campaigns to it

The channel finding that points the second model

DERCO's **own stores close 5.4% of deals below cost** against **0.5% at dealers** — an **~11× gap**. The margin leak is not spread evenly across the business; it is concentrated where DERCO controls the pricing.

Who should DERCO retain? The 14% that come back.

RFM (Recency · Frequency · Monetary) → log-scaled → KMeans. We tested $k = 2 \dots 7$ with elbow + silhouette and chose $k = 4$ — the elbow, a healthy silhouette, *and* four groups that map onto four real marketing plays.

Each segment's share of customers vs its share of total retail margin — one ordered ramp, because the segments are ranked by value

SEGMENT	CUSTOMERS	MARGIN	AVG. BUYS	PLAY
■ Champions	14.0%	25.7%	2.31	Retain
■ Big-ticket one-timers	22.8%	30.6%	1.00	Grow
■ Mainstream one-timers	32.5%	24.3%	1.00	Nurture
■ Dormant / lapsed	30.8%	19.5%	1.00	Cap spend

14% → 26%

Champions are the **only** group that buys twice, and they carry the densest margin per customer in the business.

86%

buy **exactly once** in 14 years → DERCO is **transactional, not relational**

08 · MODEL B — PREDICTING LOSS-MAKING DEALS

Flag the money-losers before they close.

Supervised classifier on **pre-sale features only** — brand, channel, comuna, price, timing. **Margin is excluded to avoid leakage** (the target is derived from it). Dummy + logistic baselines → HistGradientBoosting, stratified hold-out + 5-fold CV.

0.83

ROC-AUC, 5-fold CV mean (0.855 on the hold-out; std 0.016)

5.1×

better than reviewing deals at random

50%

of all loss-making deals caught by reviewing just the **top-10% riskiest**

Honest note: losses are only 2.18% of deals, so PR-AUC is 0.14 — modest in absolute terms, $\sim 6.5\times$ the base rate. This is a **triage aid, not an auto-reject**: a human reviews every flag.

Six numbers, one screen.

51.8%

of 2022 sales are Chinese brands



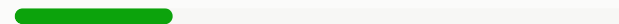
85.9%

of customers buy exactly once



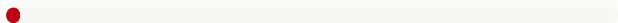
25.7%

of margin from the 14% Champions



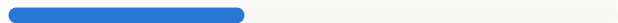
2.18%

of deals close below cost



38.8%

Suzuki's share of volume — concentration risk



50.6%

of the leak recovered for review by checking the riskiest 10% of deals



Live in `presentation/index.html`; the static twin is `presentation/dashboard.png`, written by the notebook. Both read the same `metrics.json` — no number is typed by hand.

Five findings → one decision, three funded moves.

- **China won the shelf** — 12% → 52% of sales. The core business, not a hedge.
- **Transactional, not relational** — 86% buy once; the 14% Champions drive 26% of margin.
- **~6.8 bn CLP leaks** through 2.18% loss-making deals, concentrated in the own-store channel.
- **Single-brand dependence** — Suzuki ≈ 39% of volume.
- **Geographic over-exposure** — top-10 comunas ≈ 33% of sales.

1 · Retain

Fund a **Champions-retention + one-timer-conversion** loyalty CRM.

2 · Formalise China

Treat Chinese brands as a **first-class portfolio** — pricing, stock, marketing.

3 · Plug the leak

Classifier as a **pre-approval check** on the riskiest 10%, starting with own stores.

What this analysis cannot say.

- **Synthetic money** — every CLP figure is a relative signal, not an audited financial. The direction is trustworthy; the level is not.
- **Privacy** — the customer hash is re-identifiable, so we treat it as personal data; street addresses were excluded from every model.
- **Geographic bias** — Santiago metro is over-represented, so “valuable customer” really means “*urban* customer”. A retention program must not redline rural comunas.
- **Moderate cluster quality** — silhouette 0.39. The segments are a marketing convenience, not a law of nature.
- **Human in the loop** — imbalanced target, no causal claims. The models advise; people decide. Re-check yearly.



THANK YOU

One dataset. One decision.

Retain Champions · formalise China · plug the margin leak.

- Vicente Rodríguez · Agustín Reyes · Luis-Felipe Cáceres · Baptiste Vial · UDD 2026